

Summary

Execution engineer with experience in C++, exchange gateways, shared-memory IPC, routing logic, and production trading infrastructure. Ph.D. in computer science with research in graph neural networks and explainability. Built production gateways, routing engines, risk-path optimizations, and internal UI tooling. Worked across Ops, Infra, and Research during a global execution-platform migration.

Experience

Aquatic Capital Management

Chicago, IL

Software Engineer

2020–2025

- Built and maintained exchange gateways for BAML, MSCO, and JPM.
- Improved throughput by removing basket checks from critical execution paths.
- Designed routing logic with fallback paths for reliability during broker or exchange issues.
- Implemented shared-memory IPC channels to reduce communication latency.
- Built Ops UI dashboards for routing, order status, and diagnostics.
- Actively decommissioned the legacy execution system across US/EMEA/APAC regions and onboarded the new platform.
- Collaborated with Ops, Risk, Infra, and Research teams for stability and consistency.

HERE Technologies

Chicago, IL

Senior Software Engineer

2016–2020

- Developed ML preprocessing pipelines for LiDAR perception and occupancy-grid modeling.
- Built tools for large-scale 3D data ingestion and experiment tracking.

Simplex Investments

Chicago, IL

Software Developer, HFT

2013–2016

- Built exchange protocol handlers and strategy-level routing components for low-latency order flow.
- Built backend and frontend tools that supported several arbitrage strategies.
- Developed tools for fill analysis and exchange-message debugging.

Optiver US LLC

Chicago, IL

Applications Engineer

2011–2013

- Maintained production trading systems and resolved operational incidents.
- Prototyped C++/C# based monitoring system.
- Built reporting tools.

Argonne National Laboratory

Lemont, IL

Predoctoral Appointee

2008–2011

- ZeptoOS: Helped port Linux to Blue Gene/P (BGP) Supercomputer.
- Implemented parallel benchmarks for BGP in C and MPICH2 for studying scaling characteristics.
- Designed and implemented policy management in C for HPC components.

SAP Labs

Bengaluru, India

Development Specialist

2004–2006

- Built supply-chain management features in the ABAP programming language.

Education

University of Illinois Chicago,
Doctor of Philosophy, Computer Science

May 2025

University of Illinois Chicago,
Master of Science, Computer Science

December 2008

R. V. College of Engineering,
Bachelor of Engineering, Computer Science & Engineering

June 2004

Selected Research

EEGL: Explanation Enhanced Graph Learning

 github.com/hnaik/eegl

- Ph.D. Work
- Framework using explanation-driven subgraph mining to enhance GNN expressivity

FILL

 github.com/hnaik/fill

- Exploratory code for execution and allocation; early-stage research.

Skills

Languages : C++20/23, Python

Libraries/systems : CUDA, Polars, Pandas, NumPy, PyTorch, Boost, STL, MPI

Tools : Git, CMake, GDB, Valgrind, VS Code

Trading Systems : Trading engines, gateways, risk management, simulators, UI tooling